WORKED SOLUTION

STEP II 2006, Question 1

Pure • Periodic Recurrence Relations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q1

The key idea

Because each term determines the next, the sequence repeats with period p as soon as $u_{p+1} = u_1 = 2$. So every part of (i) comes down to computing a few terms as functions of k and solving one polynomial equation. The equation for period 4 must contain the answers for periods 1 and 2 as factors (a sequence of period 1 or 2 also satisfies $u_5 = u_1$), and knowing this in advance makes the factorisation easy. Part (ii) is a one-line induction followed by the usual "let $n \to \infty$ in the recurrence" argument.

Solution

Part (i)

Write $f(u) = k - \dfrac{36}{u}$, so that $u_{n+1} = f(u_n)$. If $u_{p+1} = u_1$, then applying f repeatedly gives $u_{p+2} = u_2$, $u_{p+3} = u_3$, and so on, so $u_{n+p} = u_n$ for every n. Conversely, period p certainly needs $u_{p+1} = u_1 = 2$. So the sequence has period p *exactly* when $u_{p+1} = 2$ (with all the terms defined, that is, never 0) and no smaller period works.

(a) The sequence is constant if and only if $u_2 = 2$, that is, $k - 18 = 2$. So $k = 20$.

(b) We have $u_2 = k - 18$ and

$$u_3 = k - \frac{36}{k-18} \qquad (k \neq 18).$$

Setting $u_3 = 2$ gives $(k-2)(k-18) = 36$, so $k^2 - 20k = 0$, so $k = 0$ or $k = 20$. The value $k = 20$ gives a constant sequence, which has period 1, not 2. For $k = 0$ the sequence is $2, -18, 2, -18, \ldots$, which does have period 2. So $k = 0$.

(c) We need two more terms. With $k \neq 18$,

$$u_3 = \frac{k(k-18) - 36}{k-18} = \frac{k^2 - 18k - 36}{k-18},$$

and then, provided $k^2 - 18k - 36 \neq 0$,

$$u_4 = k - \frac{36(k-18)}{k^2-18k-36} = \frac{k^3 - 18k^2 - 72k + 648}{k^2 - 18k - 36}.$$

The condition $u_5 = 2$ says $k - 36/u_4 = 2$, that is, $u_4 = \dfrac{36}{k-2}$ (we need $k \neq 2$ here; $k = 2$ would give $36/u_4 = 0$, which is impossible). Cross-multiplying,

$$(k-2)(k^3 - 18k^2 - 72k + 648) = 36(k^2 - 18k - 36).$$

Expanding and collecting everything on the left,

$$k^4 - 20k^3 - 36k^2 + 792k - 1296 = 36k^2 - 648k - 1296,$$
$$k^4 - 20k^3 - 72k^2 + 1440k = 0.$$

Ask yourself How can we factorise a quartic like this without a lot of trial and error?

We know two roots before we start. Any sequence of period 1 or 2 also satisfies $u_5 = u_1$, so $k = 20$ and $k = 0$ must be roots. Taking out k and then $k - 20$,

$$k^4 - 20k^3 - 72k^2 + 1440k = k(k-20)(k^2 - 72).$$

So $k \in \{0, 20, 6\sqrt{2}, -6\sqrt{2}\}$. The values 0 and 20 give periods 2 and 1, so the new values are $k = \pm 6\sqrt{2}$.

Finally we check that the terms are all defined for $k = \pm 6\sqrt{2}$, using $k^2 = 72$:

- $u_2 = k - 18 \neq 0$;
- the numerator of u_3 is $k^2 - 18k - 36 = 36 - 18k$, which is non-zero since $k \neq 2$;
- the numerator of u_4 is $k(k^2 - 72) - 18k^2 + 648 = -1296 + 648 = -648 \neq 0$;
- $u_5 = 2$ by construction, and $k \neq 2$, $k \neq 18$ as required in the working.

Since $k \neq 0, 20$, the period is not 1 or 2, and a sequence with $u_5 = u_1$ whose least period is not 1 or 2 has least period exactly 4 (the least period must divide 4). So the sequence has period 4 exactly when $k = \pm 6\sqrt{2}$.

Part (ii)

Take $k = 37$. We prove $u_n \geq 2$ by induction. It holds for $n = 1$ since $u_1 = 2$. If $u_n \geq 2$, then $u_n > 0$, so $0 < \dfrac{36}{u_n} \leq 18$, and therefore

$$u_{n+1} = 37 - \frac{36}{u_n} \geq 37 - 18 = 19 \geq 2.$$

So $u_n \geq 2$ for all n, which is what we had to show. (In fact $u_n \geq 19$ for all $n \geq 2$.)

Ask yourself The equation for the limit has two roots. Which is the right one, and why can we let $n \to \infty$ in the recurrence?

Since $u_n \to \ell$ and every $u_n \geq 2$, we have $\ell \geq 2$; in particular $\ell \neq 0$, so $36/u_n \to 36/\ell$. Letting $n \to \infty$ on both sides of $u_{n+1} = 37 - 36/u_n$ (the left side also tends to ℓ) gives

$$\ell = 37 - \frac{36}{\ell} \quad \Longrightarrow \quad \ell^2 - 37\ell + 36 = 0 \quad \Longrightarrow \quad (\ell - 1)(\ell - 36) = 0.$$

The root $\ell = 1$ is ruled out by $\ell \geq 2$, so

$$\ell = 36.$$

Checks

- For $k = 6\sqrt{2} \approx 8.485$ the terms are $2, -9.515, 12.269, 5.551, 2, \ldots$, and for $k = -6\sqrt{2}$ they are $2, -26.485, -7.126, -3.433, 2, \ldots$. Both repeat after four steps.
- The factor $k^2 - 72$ has a tidy meaning: the map $u \mapsto (ku - 36)/u$ corresponds to the matrix $\begin{pmatrix} k & -36 \\ 1 & 0 \end{pmatrix}$, with trace k and determinant 36. Its eigenvalues are $6e^{\pm i\theta}$ with $k = 12\cos\theta$, and the fourth power is a multiple of the identity when $\theta = \pi/4$ or $3\pi/4$. This gives $k = \pm 6\sqrt{2}$, and $\theta = \pi/2$ gives $k = 0$ for period 2.
- With $k = 37$: $u_2 = 19$, $u_3 \approx 35.105$, $u_4 \approx 35.975$, $u_5 \approx 35.9993$, and the terms are clearly approaching 36, not 1.

Common slips

- Giving $k = 0, 20$ for (b), or all four roots for (c). The question asks for period 2 and period 4, and a constant sequence does not have period 2 in that sense.
- Taking $\ell = 1$ as well as $\ell = 36$. The bound $u_n \geq 2$ from the first half of (ii) is there to rule it out.
- Letting $n \to \infty$ without noting $\ell \neq 0$, which is needed for $36/u_n \to 36/\ell$.

Beyond STEP

The map $u \mapsto (ku - 36)/u$ is a Möbius transformation, and composing such maps corresponds to multiplying 2×2 matrices. That is why periodicity comes down to the eigenvalues having ratio a root of unity. For $k = 37$ the two fixed points 1 and 36 behave differently: 36 attracts nearby terms and 1 repels them, which you can see from $|f'(\ell)| = 36/\ell^2$. This is the start of the theory of fixed-point iteration and convergence of sequences, covered rigorously in Abbott, *Understanding Analysis*. Needham, *Visual Complex Analysis*, has a beautiful chapter on Möbius transformations and their classification.

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